

Toluidine Blue O Staining

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Purpose. Permeability and cuticle-defect assay.

Reagent / condition	Preparation or use
Leaf	0.025% TB-O in water; 10 min
Valve	0.05% TB-O + 0.01% Tween 20; 30 min
Hypocotyl	0.05% TB-O; 2 min
0.025% from 1% stock	48.75 mL D.W. + 1.25 mL TB-O 1% stock, to 50 mL
0.05% + Tween 20, 50 mL	47.5 mL D.W. + 2.5 mL TB-O 1% stock + 5 µL 100% Tween 20

1. Treat the tissue using the condition appropriate for the organ.
2. For leaves, wash in water for 5 min after staining.
3. For a rapid working workflow, briefly rinse once in D.W., place in a Petri dish containing D.W., and gently flatten with a slide glass before imaging.
4. Image by stereomicroscopy as appropriate.

Interpretation Cuticle presence and permeability are not identical properties; early embryos, for example, may have a cuticle while remaining permeable.

Source: [TB-O reference](#)